Statistical Tests Assignment

Conduct Z-test

1)A teacher claims that the mean score of students in his class is greater than 82 with a standard deviation of 20. If a sample of 81 students was selected with a mean score of 90 then check if there is enough evidence to support this claim at a 0.05 significance level.

2)An online medicine shop claims that the mean delivery time for medicines is less than 120 minutes, with a standard deviation of 30 minutes. Is there enough evidence to support this claim at a 0.05 significance level if 49 orders were examined with a mean of 100 minutes?

Conduct T-test

3) Calculate a t-test for the following data of the number of times people prefer coffee or tea in five-time intervals.

Coffee	Tea
4	3
5	8
7	6
6	4
9	7

4)A company wants to improve its sales. The previous sales data indicated that the average sale of 25 salesmen was $50 per transaction. After training, the recent data showed an average sale of $80 per transaction. If the standard deviation is $15, find the t-score. Has the training provided improved the sales?

5. Calculate correlation between the age and weight of the children.

Age	1	2	3	4	5
Weight(KG)	3	4	6	7	12

6. A professor wants to know if her introductory statistics class understands basic math well. Six students are randomly chosen from the class and given a math proficiency test. The professor wants the class to be able to score above 70 on the test. The six students get 62, 92, 75, 68, 83, and 95 scores. Can the professor have 90 per cent confidence that the mean score for the class on the test would be above 70?

7. A Sample of 16 bulbs gives a mean life of 900 hours with an S.D. of 10 hours. The manufacturer claims that the mean life of bulbs is 2000 hours. Does the sample meet the standard?

8. The blood pressure of 5 women before and after intake of a certain drug is given below: Test whether there is a significant change in blood pressure at a 1% level of significance.

Before	110	120	125	132	125
After	120	118	125	136	121

9. Conduct the ANOVA test for the following data

	program	weight_loss
1	A	2.6900916
2	A	0.7965260
3	A	1.1163717
4	A	1.7185601
5	A	2.7246234
6	A	0.6050458